

Homework 10

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(1.) Modify the code in Example 10.3 to automate Example 10.1 and obtain an answer (which is obviously $f(x) = 0$ at $x = 2$). Begin at an initial $X_0 = 3$. You should not need to modify the `grad_descent` function, only objects passed into the function, especially `func` and `grad_func`.

```
## Converged in 87 steps.
## Local minimum of 1.372959e-17 found at x = 2

##
## First three manual steps:

##      x0      x1      x2      x3
## 3.000 1.200 2.640 1.488

## f(x1) = 0.64    f(x2) = 0.4096    f(x3) = 0.262144
```

(2.) Grad student only) Modify the code in Example 10.3 to locate the minimum of the function

$$f(x_1, x_2, x_3) = e^{(x_1+5)}(x_2 + 1)^4 + x_2^2 x_3^2 + x_1^2 \sin(x_3)^2$$

```
## Stopped after 10000 iterations.

## $min_location
## [1] -5.084166e+00 -6.571351e-01  1.488928e-23
##
## $min_value
## [1] 0.01270397
```